

Sayanjib Sur

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EDUCATION

Jadavpur University <i>Master of Engineering (M.E.) in Information Technology</i>	Kolkata, India <i>Sep 2025 – Present</i>
Calcutta Institute of Engineering and Management <i>Bachelor of Technology in Information Technology</i> • SGPA: 8.46	Kolkata, India <i>Sep 2021 – June 2025</i>
Ramakrishna Mission Boys' Home High School (H.S.), Rahara <i>12th Standard, Science</i> • WBCHSE: 88.4%	Kolkata, India <i>2021</i>
Ramakrishna Mission Boys' Home High School (H.S.), Rahara <i>10th Standard</i> • WBBSE: 90.4%	Kolkata, India <i>2019</i>

TECHNICAL SKILLS

Languages: Python, Java, C, SQL
Analytics & BI: Power BI, MS Excel, DAX, Power Query, DAX Studio
Machine Learning: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
LLM & Agentic AI: LangChain, LangGraph, n8n
Web / ML Apps: Streamlit, JupyterLab
Tools: VS Code, Git, GitHub, MySQL Workbench
Cloud & APIs: Google Gemini API, REST APIs, Model Serving (Local/Cloud)

PROJECTS

- SLM Benchmarking Pipeline for Structured Questionnaire Reasoning** | *accepted at COMSYS 2026*
- Designed a cluster-aware prompting pipeline partitioning 5 SLMs (1.5B–7B) into context-length groups (2K–4K, 8K, 16K+) with distinct zero-shot, one-shot, and self-augmented prompt families calibrated per cluster.
 - Conducted **10,800 inference runs per model** across 6 task types, 6 serialization formats (HTML, JSON, Markdown, Turtle, TXT, XML), and 5 diverse questionnaire datasets using local Ollama deployment on Google Colab Pro (A100).
 - Introduced an MCQ-aware rescoring layer to recover valid option-letter predictions; best overall accuracy: **Vicuna-7B (35%)**; multi-hop inference and rule-based querying collapsed below **6%** across all models, exposing a hard reasoning ceiling in current SLMs.
- SLM-Based Multi-Agent Frameworks for Mathematical Reasoning** | Submitted at ICCUBEA 2026
- Designed two complementary multi-agent methodologies evaluated on the **SVAMP benchmark** across 5 SLMs (1.5B–7B): **Pipeline Decomposition** (sequential Planner → Solver → Verifier with 3 controllable communication modes: natural, constrained, structured) and **Diverse Ensemble Reasoning** (parallel CoT, Formula-First, and Backward agents aggregated via majority vote).
 - Best pipeline accuracy: **DeepSeek-R1-Distill-Qwen-1.5B (67%)** with natural communication; structured communication outperformed natural communication on verbose models by reducing noise entering the Solver; Planner-based distractor identification was the critical factor on SVAMP's deliberately misleading number sets.
 - Diverse Ensemble achieved **LLaMA 3.2-3B at 81% accuracy** (vs. 77% CoT baseline), with a disagreement rate of 61% across strategies; introduced **NATE** (Normalized Accuracy per Thousand Tokens) as an efficiency metric for fair cross-model comparison under token-budget constraints. [GitHub](#)

Fake News Detection using Machine Learning | *Python, NLP, TF-IDF, Scikit-learn*

[GitHub](#)

- Processed 20,000+ news articles with NLP (tokenization, lemmatization, TF-IDF).
- Compared 4 ML models: Logistic Regression (**97.4%** accuracy), Naive Bayes (**94.1%**), SVM (**98.7%** accuracy, **99.2%** recall), Random Forest (**96.7%**).
- SVM with Linear Kernel outperformed ensembles, proving effectiveness of linear models in text classification.

EXPERIENCE

Research Intern — IIT Patna (AI-ML-NLP Lab)

Feb 2026 – May 2026

- Working under Dr. Sriparna Saha on NLP for low-resource Indic languages; hands-on experience with multilingual models including Sarvam, Airavata, OpenHathi, IndicBART, Gemma, and LLaMA families.
- Built end-to-end pipelines to scrape, clean, and preprocess web data for low-resource languages, constructing a corpus of **15,000+ samples**.

Teacher & Founder — SS Classes

2023 – Present

- Founded a coaching initiative for **50+ undergrad students** (MAKAUT/RCCIIT); taught Engineering Mathematics and Automata Theory, achieving **15–18 backlog clearances** and **15+ students with SGPA ≥ 8.0** .

Generative AI Developer Intern — AI Wallah

May 2025 – June 2025

- Built LLM-powered chatbots and a YouTube summarization tool using fine-tuning and Google Gemini API.

PUBLICATIONS

1. Sur, S., and Bhaumik, P. “Evaluating SLMs on Understanding Structured Frameworks using Cluster Aware Prompting.” *Proceedings of COMSYS 2026*. [Accepted].
2. Sur, S., Sil Sarma, A., and Singh, P. K. “Structured Communication and Ensemble Diversity in SLM based Multi-Agent Systems: A Comparative Study.” Submitted to *IEEE International Conference on Computing, Communication, Control and Automation (ICCUBE 2026)*.
3. Sur, S., Ghosh, D., and Singh, P. K. “Small Language Models: A Survey of Architecture, Scaling, Applications, and Deployment (2022–2026).” *Manuscript in Preparation*. [Ongoing].

LANGUAGES

Bengali, English, and Hindi

ACHIEVEMENTS

GATE CS/IT 2025 Qualified | Secured 96.27 percentile (Top 3.73% nationally)

UGC NET Dec 2025 Qualified (PhD Only) | Secured 97.45 percentile (Top 2.5% nationally)

LEADERSHIP / EXTRACURRICULAR

Celsius 2024 — Hackathon Co-Head

2024

Calcutta Institute of Engineering and Management

- Coordinated a hackathon with 20+ teams, managing logistics, planning, and execution.
- Ensured smooth scheduling, participant coordination, and competitive coding environment.